Resonant conversion of dark-photon fields in a plasma with a noise-free continuum kinetic method: bounds on nonlinear suppression and the role of collisions

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Particle-in-cell simulations have been argued to show that nonlinear plasma dynamics cuts off the resonant conversion of dark-photon dark matter into plasma oscillations in the early Universe (Hook, Huang and Shalaby). We test this with a noise-free Hermite–Fourier Vlasov solver for the nonrelativistic Maxwell–Proca–Vlasov system with kinetic mixing, in one spatial and three velocity dimensions. The method is limited by a truncation instability of the asymmetrically weighted Hermite basis, which bounds the certified windows (the times over which refinement, positivity and energy-ledger checks pass) to 1000, 892 and 791.5 in units of ω_{pe}^{-1} at drive strengths $v_q/v_{te} = 0.03$, 0.05 and 0.07; the alternative bases and closures we tested did not remove it. Inside the windows, at $v_q/v_{te} = 0.07$ the energy transfer follows the linear resonant law to 0.68 % with no resolved frequency detuning (bound $6.5 \times 10^{-6}\ \omega_{pe}$). A finite dark reservoir and a finite coherence time of the drive leave the linear prediction unchanged to 0.81 % of the initial reservoir energy and 1.7 % of the ensemble work. With collisions, the kinetic absorption equals the linear Drude law with $\nu_{\rm eff} = (\nu_{ei}/2)(1+m_e/m_i)$: the late rate is within 0.4 % of the steady-state power at $\nu_{ei} = 3\times10^{-2}\omega_{pe}$, and the work scales as $(v_q/v_{te})^p$ with $p = 1.99 \pm 0.10$. At the one drive that matches a published particle-in-cell run ($v_q/v_{te} = 0.03$), our electron $k \neq 0$ energy is 0.1 initial thermal energies at $\omega_{pe}t = 500$ and 2.5 at $\omega_{pe}t = 1000$, against about 50 reported there, and the mean-field energy stays at 0.96 of the linear value: we do not reproduce the saturation at the matched drive. The model is nonrelativistic, whereas the coherent quiver speed reaches $0.47c$ at that drive and $0.87c$ at the strongest; relativistic detuning, absent here, is an untested candidate for the difference. These are bounds in a restricted model, not a resolution of the cosmological question: they hold in one dimension for $v_q/v_{te} \leq 0.07$ and $\omega_{pe}t \lesssim 790$ at the strongest drive, before the oscillating two-stream cascade develops.

I. INTRODUCTION

A dark photon with kinetic mixing χ with the ordinary photon and a mass $m_{A'}$ is a well-motivated candidate for dark matter [1]. In the early Universe the dark-photon field oscillates coherently, and when $m_{A'}$ equals the plasma frequency of the ionized medium the field resonantly converts into longitudinal plasma oscillations, depositing energy in the plasma [2–4]. Cosmological constraints on dark-photon dark matter have been derived from the resonant transfer of a substantial part of the dark-photon energy density into the plasma, which heats it.

Hook, Huang and Shalaby (HHS) argued that the resonant transfer saturates because of plasma nonlinearities [5]. In their account, large-amplitude $k \simeq 0$ Langmuir waves, driven by the dark photon, excite higher-k Langmuir and ion-acoustic waves through the ponderomotive force; the resulting density inhomogeneities detune the resonance and limit the deposited energy to roughly the electron thermal energy, which would weaken the constraints by several orders of magnitude. Their evidence comes from one-dimensional particle-in-cell (PIC) simulations (SHARP, with a relativistic Vay pusher) of an electron–ion plasma with $m_i/m_e = 1836$ and a prescribed uniform dark-photon drive: resonant runs at $v_q^D/v_{th}^e = 0.03$ and 10^{-3} (their Fig. 2) and Landau–Zener runs (their Fig. 3), published as Phys. Rev. Lett. 137, 071004. They quote a saturation of the energy transfer at roughly the electron thermal energy for weak drives and up to about 10^2 times it for strong ones.

A PIC calculation carries sampling noise, which seeds the very instabilities that the argument relies on. This paper asks a narrow question and answers it with a different numerical method: in a model with the same physical ingredients, when does the energy transfer from a resonant uniform drive depart from the linear resonant law, and how large is the departure that can be resolved? We write $v_q/v_{te} \equiv E_0/(v_{te}\omega_{pe})$ for the drive strength, with E_0 the drive amplitude, v_{te} the electron thermal speed and ω_{pe} the electron plasma frequency (Sec. II D). We use a noise-free continuum kinetic solver whose seeds are deterministic and whose work ledger closes to solver tolerance, so that any departure from the linear law can be attributed. We do not reproduce the HHS simulations: our model is nonrelativistic, the loading is deterministic, and the box and parameters differ in detail; Sec. VII A states which of our numbers are comparable to theirs and which are not.

The work consists of a methods part and a bounds part. The methods part (Secs. II–IV) describes the model, the Hermite–Fourier discretization, the verification suite, and the failure mode that limits the method. The results part addresses two research questions: (i) does a finite dark-photon reservoir or a finite coherence

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time of the drive produce detuning or saturation of the energy transfer at the strongest drives we can certify (Sec. V)? and (ii) with collisions present, does the sustained absorption obey the linear drag law, and what is missing before a cosmological statement (Sec. VI)? Failed and amended checks are reported where they affect a claim.

II. MODEL

A. Canonical Maxwell–Proca–Vlasov system

Each species s obeys the collisionless Vlasov equation in the combined force field,

$$\partial_t f_s + \mathbf{v}\cdot\nabla f_s + \frac{q_s}{m_s}\Big[\mathbf{E} + \eta\mathbf{E}_D + \mathbf{E}_{\rm drive} + \mathbf{v}\times(\mathbf{B}+\eta\mathbf{B}_D)\Big]\cdot\nabla_v f_s = 0, \quad (1)$$

with ordinary fields $\partial_t\mathbf{E} = c^2\nabla\times\mathbf{B} - \mathbf{J}/\epsilon_0$, $\partial_t\mathbf{B} = -\nabla\times\mathbf{E}$ and canonical Proca fields

$$\partial_t\mathbf{E}_D = c^2\nabla\times\mathbf{B}_D + \Omega_D^2\mathbf{A}_D - \eta\mathbf{J}/\epsilon_0, \quad (2)$$

$$\partial_t\mathbf{B}_D = -\nabla\times\mathbf{E}_D, \quad \partial_t\mathbf{A}_D = -\mathbf{E}_D - \nabla\phi_D, \quad (3)$$

$$\partial_t\phi_D = -c^2\nabla\cdot\mathbf{A}_D, \quad (4)$$

with constraints $\nabla\cdot\mathbf{E} = \rho/\epsilon_0$ and $\nabla\cdot\mathbf{E}_D + \Omega_D^2\phi_D/c^2 = \eta\rho/\epsilon_0$. The ordinary current $\mathbf{J}$ sources both laws; the dark field acts back on the particles with strength η. Three modes are used: *ordinary* (no dark coupling), *prescribed drive* (a uniform $\mathbf{E}_{\rm drive} = \mathbf{E}_0\cos(\omega t+\varphi)$ and an external-work ledger, no dark reservoir; the setting of HHS), and *self-consistent* (evolved Proca fields with a finite energy reservoir).

B. Mixing map

For a kinetic mixing $-\frac{\chi}{2}F_{\mu\nu}F_D^{\mu\nu}$ and a mass term on the dark field only, removing the mixing by a non-orthogonal shear gives the canonical coupling

$$\eta = \frac{\chi}{\sqrt{1-\chi^2}}, \qquad m_D = \frac{m_0}{\sqrt{1-\chi^2}}, \quad (5)$$

where m_0 is the bare mass and Ω_D is the dark-photon mass m_D expressed as a frequency. We checked this map against Sec. 1.1 of Ref. [1] (their $\varepsilon \equiv \chi$): the dark photon couples to the ordinary current with strength $e\varepsilon/\sqrt{1-\varepsilon^2}$. The sign convention differs by $\mathbf{A}_D \to -\mathbf{A}_D$, under which the model is symmetric in η. The model is not the massless "millicharged" choice of the same reference.

C. Energy and work ledger

In the parent normalization velocities are in units of c, $\Omega_0 \equiv \Omega_{c,0}$ and the species mass is $m_s = \Omega_0/\Omega_{cs}$, which is required for the energy theorem. With $\langle\cdot\rangle$ the box average the field energies and powers are

$$U_\gamma = \tfrac{1}{2}\Omega_0^2\langle E^2 + B^2\rangle, \quad (6)$$

$$U_D = \tfrac{1}{2}\Omega_0^2\big\langle E_D^2 + B_D^2 + \Omega_D^2(A_D^2+\phi_D^2)\big\rangle, \quad (7)$$

$$P_\gamma = \Omega_0\langle\mathbf{J}\cdot\mathbf{E}\rangle, \quad P_D = \Omega_0\eta\langle\mathbf{J}\cdot\mathbf{E}_D\rangle, \quad (8)$$

$$P_{\rm ext} = \Omega_0\bar{\mathbf{J}}\cdot\mathbf{E}_{\rm drive}. \quad (9)$$

The works W_γ, W_D, $W_{\rm ext}$ are integrated as extra ODE states with the same Runge–Kutta stages as the fields, so that $K - K_0 = W_\gamma + W_D + W_{\rm ext}$, $U_\gamma - U_{\gamma 0} = -W_\gamma$ and $U_D - U_{D0} = -W_D$ hold to solver tolerance. The Proca energy flux is $\epsilon_0 c^2\mathbf{E}_D\times\mathbf{B}_D + \epsilon_0\Omega_D^2\phi_D\mathbf{A}_D$; omitting the $\phi_D\mathbf{A}_D$ term breaks the flux budget (Sec. IV). For a homogeneous plasma under the uniform drive the mean field obeys a driven oscillator at the plasma frequency $\omega_L = \sqrt{1+m_e/m_i}\,\omega_{pe}$, whose solution is the *linear resonant law*: $\dot{\bar{E}} = -\bar{J}$, $\dot{\bar{J}} = \omega_L^2(\bar{E} + E_0\cos\omega t)$, $\dot{W} = \bar{J}E_0\cos\omega t$. This law is exact in the homogeneous discrete Hermite system because only Hermite orders 0–1 enter (a homogeneous check, agreement to 10^{-9} for $\omega_{pe}t \le 1000$ even at quiver speeds of $15\,v_{te}$). With fluctuations present it is the baseline against which the transfer is measured, and the departure from it contains the $\langle\delta n\,\delta E\rangle$ back-reaction term of the mean-field identity.

D. Units and the HHS-like set-up

All simulations use $c = \omega_{pe} = n = 1$ for the electrons, $m_i/m_e = 1836$, $T_i = T_e$, $v_{te}^2 = 10^{-3}c^2$, box length $L = 40\,c/\omega_{pe}$ and a prescribed drive at $\omega = \omega_L$ unless stated. We call $v_q/v_{te} \equiv E_0/(v_{te}\omega_{pe})$ the effective initial force (the quiver speed grows secularly with time at resonance). Energies are quoted in units of nT_eV. HHS use $v_q^D/v_{th}^e = 0.03$ and 10^{-3} for their resonant runs (PRL Fig. 2), $m_i/m_e = 1836$, $k_BT_e = k_BT_i = 10^{-3}m_ec^2$ ($v_{th}^e = 3.1\times10^{-2}c$), $L = 40\,c/\omega_{pe}$ with 1000 cells of $0.04\,c/\omega_{pe}$ and a prescribed spatially uniform drive $A_0\cos\omega t$ (Supplemental Material, Sec. II); we use the same v_{te}, mass ratio, temperature ratio, box length and v_q definition, but not the loading (their particles start equally spaced, ions and electrons colocated; ours is a deterministic spectral seed) or the relativistic pusher. Runs here are *HHS-inspired*, not reproductions.

III. METHOD

A. Hermite–Fourier discretization

The velocity dependence is expanded in Cartesian asymmetrically weighted (AW) Hermite functions with independent orders (N_n, N_m, N_p), widths α_s and centres u_s; space is Fourier, with the strict 2/3 dealiasing mask of the parent (which retains $3|m| < N_x$, so every grid size is alias-free for quadratic products). A

single call to the SPECTRAX Hermite–Fourier kernel per right-hand-side evaluation receives the physical force fields $\mathbf{E}+\eta\mathbf{E}_D+\mathbf{E}_{\rm drive}$ and $\mathbf{B}+\eta\mathbf{B}_D$, and the current is evaluated once for both Ampère laws. Time integration is Dopri8 with a PID controller (Diffrax). The kinetic-energy ledger closes only when every forced direction has Hermite order at least 3. The parent is SPECTRAX [6]; the studies reported here were run at the parent commits recorded in each record, all on an unmerged integration branch of open parent changes, not a release (see Data and code availability).

B. Pump frame and basis remaps

In the pump frame the state carries the centres u_s and widths α_s. Each centre follows the total uniform force, $\dot{u}_s = (q/m)_s(\bar{\mathbf{E}} + u_s \times \bar{\mathbf{B}})$, with the box averages including $\eta\mathbf{E}_D$ and the drive; the parent kernel then uses $\mathbf{E}-\bar{\mathbf{E}}$. This is an exact change of variables: the coherent $k=0$ quiver sits in $u_s(t)$ rather than in high Hermite orders. Widths and residual drifts change only between integration segments, through the parent's exact lower-triangular remap (every moment below the truncation is preserved); triggers fire on $|U-u|/a > 0.3$ or $|a'/a-1| > 0.05$, with caps $|\Delta u| \le a'$, $a/a' \le 1.1$, $a' \ge 1.1\sigma$.

C. Closures

Truncation needs a closure. The parent closure is a hypercollision $\propto n(n-1)(n-2)$ that leaves Hermite orders 0–2 unchanged. We use a field-scaled form, implemented in the SPECTRAX integration branch,

$$\nu_s = c\left|\frac{q}{m}\right|_s \frac{\sqrt{2N_n}\,\max|E-\langle E\rangle|}{a_s} \times \frac{n(n-1)(n-2)}{(N_n-1)(N_n-2)(N_n-3)}, \quad (10)$$

with strength c (density, momentum and energy rows untouched; a Lorentz-force-scaled variant exists for magnetized runs and is not needed here). For $c=64$ the closure rate is the stiff eigenvalue of the system: the Jacobian's spectral radius equals the closure rate ν_e (real and negative) to the printed precision, so the explicit step is bounded by $5.17/\nu_e$. We therefore integrate the diagonal closure damping exactly with a Lawson integrating factor [7] wrapped around Dopri8 (integrating-factor and exponential Runge–Kutta methods are reviewed in Ref. [8], and for kinetic equations in Ref. [9]). Lawson transforms $y' = f(t,y)$ with a matrix A by $z = e^{-tA}y$ and advances z with an explicit Runge–Kutta method, so that the stiff part of the Jacobian, $e^{-tA}(\partial f/\partial y)e^{tA} - A$, is removed from the step-size restriction; A may be re-evaluated at each step from the local Jacobian (his Algorithm II) [7]. Here $A = -\mathrm{diag}(r)$ is the closure rate frozen at the step start, so with $w = e^{r(t-t_0)}y$ the split is an identity within the step; streaming, the force, the fields and the variation of ν over the step stay explicit. Lawson shows that the construction is A-stable for the linear test problem if the exponential is approximated stably, is stable for $y' = By + u(t,y)$ when the Lipschitz constant of u is small compared with the spectral radius of B, and keeps the order of the underlying Runge–Kutta method when $\exp(hA)$ is approximated to at least that order [7]; here the diagonal exponential is exact. These statements hold for $h \to 0$ at fixed A. For semilinear problems with large A, Hochbruck and Ostermann show that the stiff order of such methods can be lower than the classical one (the Lawson–Runge scheme is only first-order accurate under their stiff-order theory, Example 2.32 and Theorem 2.17 of Ref. [8]), so we do not claim that the order of Dopri8 is retained when $rh \gg 1$; we rely on the measured agreement with the unfactored integration instead. For $c=64$ the factor reduces the step count by a factor 5.0 to $t=500$ with ΔK_e and $W_{\rm ext}$ agreeing with the Dopri8 record to 5×10^{-10}; for $c=4$, where the closure is not stiff, it gives 0.96 (no gain). An exact Hermite-streaming factor was also built and rejected: it gave no step reduction and cost 2.7 times more per step. Table II lists these and other negative results.

D. Collisions

Collisions use the Lenard–Bernstein–Dougherty (LBO) operator [10], $C[f] = \nu\,\partial_{\mathbf{v}}\cdot\big[(\mathbf{v}-\mathbf{u})f + (T/m)\,\partial_{\mathbf{v}}f\big]$, with a velocity-independent frequency ν, an isotropic, velocity-independent diffusion coefficient, and $\mathbf{u}$ and T the flow and temperature moments of f itself (his Eqs. 7–11 and 14). Dougherty shows that it conserves number, momentum and energy and that its only equilibrium is the Maxwellian, and he states its limits: it is a single-species model, accurate for most particles and wrong for the high-energy tail, and the multispecies case is "similar in principle but more cumbersome". We use the conservative Hermite discretization of Hakim et al. [11] with the multispecies forms of Francisquez et al. [12], in two cross-species conventions: LBO-EM (the parent default, the form of Francisquez et al. Eqs. 2.18–2.20: momentum rate equal to the Morse rate and temperature-difference rate equal to twice the Morse rate) and LBO-G with $\beta = 0$ (both rates equal to the Morse rates). The collision frequencies are velocity independent, $\nu_{ee} = \nu_{ei}$, $\nu_{ii} = \nu_{ei}\sqrt{m_e/m_i}$, $m_e\nu_{ei} = m_i\nu_{ie}$; the constancy of ν follows Dougherty's model and is a limitation for the cosmological case (Sec. VI). Collisions do no net work and conserve mass, momentum and energy in the homogeneous controls to round-off.

IV. VERIFICATION AND LIMITS

A. Verification

Table I and Fig. 1 summarize the checks. The linear Landau roots of the Maxwell–Proca–Vlasov system (with $k\lambda_{De} = 0.2$–0.7, $\eta = 0.3$, $\Omega_D = \omega_{pe}$) agree with SciPy Faddeeva roots of $D_L = (Q - \Omega_D^2)(1 + \chi) + \eta^2 Q\chi$ to 1.3×10^{-4} across 32 runs, but at $k\lambda_{De} = 0.2$ the damping rate ($\sim 5 \times 10^{-5}$) is not resolved inside the Hermite-limited fit window; only the frequency is validated there. The oscillating two-stream instability (OTSI) that sets the high-k cascade is recovered from the Hermite Floquet analysis against the kinetic dipole-pump dispersion relation of Silin [13] (his Sec. 4, Eqs. 4.4–4.8). For a uniform pump with electron excursion r the perturbation splits into sidebands $\omega + l\omega_0$ coupled by Bessel factors $J_l(kr)$, with Maxwellian susceptibilities χ_s for each species. We evaluate it as the determinant of the sideband matrix $\mathrm{diag}(1+\chi_i)+B\,\mathrm{diag}(\chi_e)B^T$, $B_{lj} = J_{l-j}(kr)$, $|l| \le 4$, with unpumped ions, and compare its purely growing root with the Floquet exponent at 22 unstable points: the worst difference is 1.3 %. In Silin's analysis (cold-plasma Eqs. 3.24–3.35 and the kinetic generalization of Sec. 4) the growth is purely exponential for a pump below the Langmuir frequency, $\omega_0^2 < \omega_{Le}^2 + \omega_{Li}^2$, and oscillatory above it; the rate has a weak-field scaling $(m_e/m_i)^{1/3}\omega_{pe}(kr)^{2/3}$ and a maximum of order $(m_e/m_i)^{1/3}\omega_{pe}$ for $kr \sim 1$ near a harmonic resonance. Nishikawa's coupled-mode theory [14, 15] gives the same classification (non-oscillatory growth for $\omega_0 < \omega_2$, oscillatory for $\omega_0 > \omega_2$, with ω_2 the frequency of the pumped Langmuir wave) and the cube-root dependence on pump power, $y_m = (K/2\omega_2)^{1/3}$ (I, Eq. 40a), but for a fluid plasma with phenomenological damping and a pump energy small compared with the thermal energy (II, Sec. 2). We therefore use it for the classification and the scaling, not as a quantitative reference. With fixed ions every rate vanishes, so the instability needs mobile ions. The relation is known physics; what is new here is the regime (resonant pump with a secularly growing excursion and $T_i = T_e$). Further checks, all in the repository: cold coupled oscillators to 10^{-11}, Proca vacuum and Yukawa limits, the η-sign symmetry, the exact mean pump, a Weibel and filamentation set against an independent kinetic dispersion relation including Proca coupling (within 10^{-3}), magnetized rotation and cold electromagnetic eigenmodes (within 10^{-3}), and Dougherty collisional relaxation against the exact rates (to 10^{-13}). Independent discretizations: a NumPy semi-Lagrangian grid solver (and its Vlasov–Ampere–Proca extension) agrees with the Hermite echo amplitude to 2×10^{-5}; the Dark-JAX-in-Cell PIC code [16] approaches the Hermite Landau fit as cells and particles increase, with a remaining difference of order 1 % in damping set by PIC noise (Fig. 1c); this is not a claim of PIC convergence. GPU and CPU agree to 9.7×10^{-9} on one case only. Gradients of converged dark observables are verified (Table I); reverse mode through Diffrax's `RecursiveCheckpointAdjoint` was found wrong (2–15 %) for many constant-step saves, and `DirectAdjoint` is used.

B. The Hermite truncation instability and the certified windows

The method has one limit that matters for the rest of the paper. In a non-uniform field the truncated AW Galerkin operator $-v\partial_x + (q/m)E\partial_v$ is not anti-self-adjoint; its frozen-field spectrum has growing modes in the top third of the Hermite ladder, with a rate that rises with N_n (Fig. 2a; at $K = 5$, $N_n = 128$, $E_1 = 10^{-2}$ the largest growth rate is 1.6, against $\le 8 \times 10^{-15}$ for the symmetric basis, which is neutral to round-off). This is a numerical instability of the basis, not physics. It is distinct from the physical OTSI, which the Hermite model reproduces and which amplifies finite-k fields strongly at resonance (the Hermite model understates the growth: Nn 32/64/128 agree with each other and end 1.3–4 times below the exact solution). The AW instability is excited by the large-amplitude pump and ends the simulation by loss of positivity or by the step budget.

A *certified window* is defined by checks declared before the runs: refinement pairs (N_x, N_n, closure strength) must agree to 10 % in ΔK_e and W_{ext}, the electron distribution must stay positive ($K_s > 0$ and $k = 0\ T_x > 0$), the solver must reach the time, and the energy ledger must close. The window is the minimum of these times. It is therefore a lower bound on where the Hermite results can be trusted, not a statement about physics.

Alternatives were tested and rejected, all under predeclared gates (Table II). Higher resolution on GPU ($N_x \in \{64, 128\}$, $N_n \in \{128, 256\}$, $v_q/v_{te} = 0.1$ and 0.05) failed the declared window-extension check and ended every run inside the instability at 328–595 ($v_q/v_{te} = 0.1$) and 440–751.5 ($v_q/v_{te} = 0.05$): the pairs agree to the end of the shorter run, so this is a failure of the resolution route rather than a disagreement. In a frozen field no closure strength up to 1024 brings the AW growth below 10^{-3} at peak field 4×10^{-3} (39/39 points fail), leaving a floor that grows as $K^{0.15}N_n^{0.30}E^{1.26}$, so a resolution-consistent closure strength does not exist for this closure family. A stable Galerkin form of Dai [17] is unstable in the AW coordinates in float64 at $N_n \ge 64$ (the AW-to-orthonormal change of basis has condition $\sim 2^N$) but stable and accurate against the kinetic OTSI relation in symmetric Hermite functions of width $a/\sqrt{2}$; the symmetric-weight basis itself, and the anti-symmetric square-root form of Issan et al. [18, 19], are linearly neutral to round-off and accurate against the same relation, but were rejected under their declared nonlinear $v_q/v_{te} = 0.1$ gates (symmetric weight: every run blows up at $t = 637$–738 and the refinement pair agrees only to 634; square-root form: departure from the grid reference at 391 and 498, earlier than the symmetric weight).

TABLE I. Verification summary. Record files are listed under Data and code availability.

Check	Result (record)
Linear Landau roots, ordinary and dark, $k\lambda_{De} = 0.2$–0.7	worst relative error 1.3×10^{-4} vs. SciPy Faddeeva roots; ledger defect $\leq 3 \times 10^{-18}$
OTSI growth vs. Silin kinetic relation (sideband determinant)	22 unstable points, worst difference 1.3%
Dark-JAX-in-Cell PIC, Landau $k\lambda_{De} = 0.5$	rms $\Delta \ln \lvert E_k \rvert$ falls 0.22→0.13 from 32 to 128 cells; PIC noise limits agreement to ∼1%
GPU vs. CPU, one case ($T = 100$)	max relative difference 9.7×10^{-9} (adaptive trajectories)
AD vs. finite differences	reverse = forward to 2×10^{-13}, analytic to 5×10^{-10} (19 derivatives)

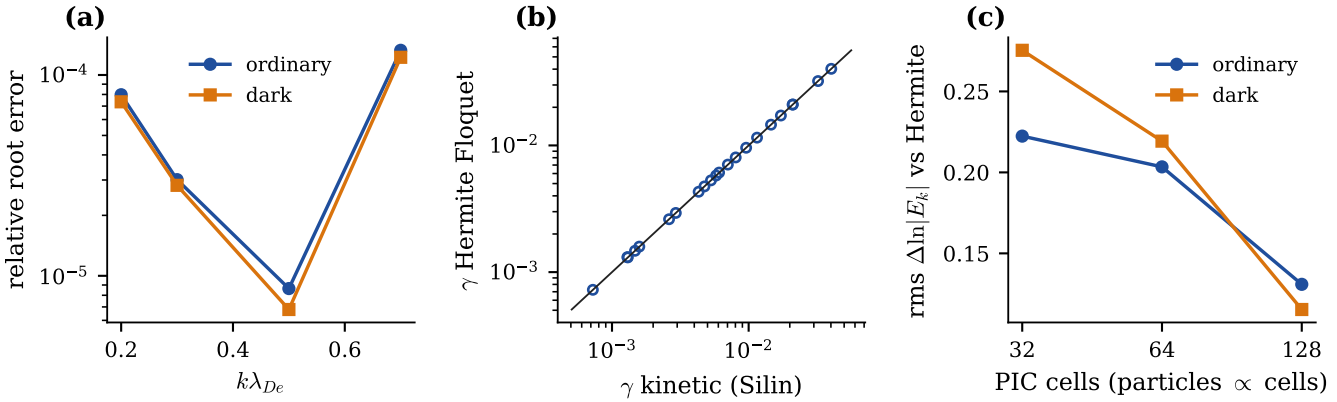


FIG. 1. Verification. (a) Relative error of the fitted complex frequency against the independent kinetic root, ordinary and dark ($\eta = 0.3$), $N_n = 128$, $N_x = 8$. (b) Hermite Floquet growth rate against the kinetic dipole-pump relation of Silin (sideband determinant, see text). (c) RMS difference of $\ln |E_k|$ between Dark-JAX-in-Cell and Hermite on $2 \leq \omega_{pe} t \leq 12$ for Landau damping, for increasing cells and particles.

Production therefore stays with the AW basis and the field-scaled closure, and the windows below are its windows.

V. FINITE RESERVOIR AND FINITE COHERENCE

A. Design

Two idealizations of the prescribed-drive calculation are removed. First, in the self-consistent mode the dark photon is a free $k = 0$ Proca mode with $\Omega_D = \omega_L$ whose energy reservoir U_{D0} is finite and is depleted by the transfer; we use $u = U_{D0}/(nT_eV) = 0.1, 1, 10$ with η chosen so that the initial effective force matches the prescribed drive. Second, the dark field has a finite coherence time: the prescribed drive is $E_0 \cos(\omega_L t + \Phi(t))$ with Φ a phase random walk of independent Gaussian frequency samples, giving a Lorentzian line of half-width $1/\tau_c$ and no phase resets. The comparison is always with the exact linear law for the same drive (and, for the reservoir, the linear coupled oscillator $(\bar{E}, \bar{J}, E_D, A_D)$, whose matrix exponential agrees with an ODE solver to 6.6×10^{-11}).

B. Windows at each drive

Table III gives the windows: 1000, 892 and 791.5 for $v_q/v_{te} = 0.03$, 0.05 and 0.07. The windows are set by positivity loss of the $N_x = 64$ members, that is, by the AW truncation instability reaching the resolved range, not by disagreement of the refinement pairs. The strongest certified drive is $v_q/v_{te} = 0.07$ with window 791.5. Drives above that, including $v_q/v_{te} = 0.1$, have no window beyond about 760 (Sec. IV).

C. Linear transfer and the seed back-reaction

At $v_q/v_{te} = 0.07$ the work $W_{\rm ext}$ follows the exact linear resonant law over the whole window (Fig. 3a): the maximum of $|W_{\rm ext}/W_{\rm lin} - 1|$ over four resolutions is 0.68 %, the four members agree on it to 2×10^{-4}, and at the window end $W_{\rm ext} = 381.3$ against $W_{\rm lin} = 384.0$ (in nT_eV). The mean-field frequency shift, measured as the slope of the lock-in phase difference, is at most $6.5 \times 10^{-6}\ \omega_{pe}$ (the declared resolution threshold was $10^{-3}\omega_L$; not resolved). No saturation and no detuning are resolved. The departure rule (a 10 % departure in the coarsest member confirmed at 5 % in all refinements) is not met.

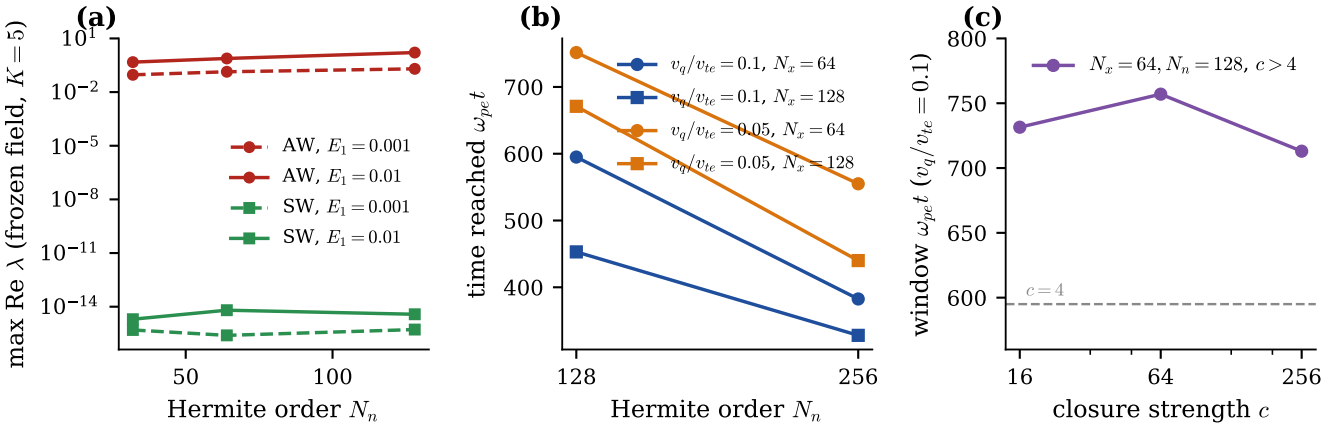


FIG. 2. The limit of the method. (a) Largest real part of the frozen-field spectrum for the AW basis (grows with N_n) and the symmetric-weight basis (round-off), $K = 5$. (b) Time reached by AW runs with closure $c = 4$ at $v_q/v_{te} = 0.1$ and 0.05 for $N_x = 64, 128$ and $N_n = 128, 256$, all ended at the step budget inside the instability: more resolution ends the run sooner. (c) At $v_q/v_{te} = 0.1$, $N_x = 64$, $N_n = 128$ a stronger fixed closure extends the window from 595 to a ceiling of 713–757.

TABLE II. Negative and amended results that bound the claims. "Declared" means the check was committed before the run and not amended.

Study	Check	Outcome	Consequence
GPU resolution extension	Longer window at higher resolution, $v_q/v_{te} = 0.1$, 0.05	Failed; every run ends at the step budget inside the instability	No new window; a stable basis is needed
Closure-strength route	AW growth below 10^{-3} in a frozen field for some closure strength	Failed; a floor remains	Scaled-closure option not built
Stiffness vs. instability	Which eigenvalue limits the step	Stiff eigenvalue is the closure; the instability (max Re $\lambda = 2$–3) is not	Integrator is not the limit
Lawson factor (closure)	Step reduction at $c = 4$	Failed as declared (no gain)	Useful for $c = 64$ only
Lawson factor (streaming)	Step reduction with an exact streaming factor	Failed	Rejected
Dai / symmetric weight / square root	Declared nonlinear gates at $v_q/v_{te} = 0.1$	Failed (Sec. IV)	AW basis with closure stays
Finite coherence, first version	Declared phase variance	Wrong: realized τ_c was 2× nominal	Corrected version declared and used
Seed deficit hypothesis	"Homogeneous truncation effect"	Rejected	Deficit is seed back-reaction
Collisional steady-state check	Window longer than the record	Declaration defect; declared follow-up used, deviation from P_{ss} 3×10^{-5}	Follow-up result used
Collisions, 5 % agreement	Kinetic equals linear within 5 %	30 of 36 pass; six detuned cases at $\nu_{ei} = 10^{-3}$ fail	Seed effect; seed-scale follow-up passed
Earlier collision pilot	Restart-chunk length; ion heating	Failed as declared (T_i responds to the halved exchange)	Recorded, not amended
Collision-form controls	Several declared checks	Mis-specified or at round-off level	Recorded; ledger closes to 10^{-17}

The residual deficit is not a drive effect. At the three drives it is the same to three digits (it grows as $-10^{-8}\,t^2$) until the OTSI stage, ending at -1.07, -0.88 and -0.68 % at the window ends. A seed-scale test at $v_q/v_{te} = 0.07$ (the initial density fluctuations multiplied by s) shows the deficit scales as s^2 (Table IV, Fig. 3b): the ratio of $s = 0.1$ to $s = 1$ is 0.0100 and the unseeded run gives $2.3{\times}10^{-9}$ at $\omega_{pe}t = 400$. The deficit is therefore the back-reaction of the quiver-advected seed modulation on the mean field (the $\langle \delta n\, \delta E \rangle$ term of the mean-field identity), set by the imposed fluctuation level, not by the pump. A physical fluctuation level different from 5×10^{-4} would rescale it. Our earlier hypothesis that it is a truncation effect of the seedless homogeneous system was tested and

TABLE III. Certified windows for the strong-drive set-up with closure $c = 64$ integrated exactly. Members: $(N_x, N_n) = (32, 64), (64, 64), (32, 128), (64, 128)$ and $(32, 64)$ at $c = 32$. The window is the minimum of refinement-pair times, positivity times, times reached and the cap $T = 1000$; the grid column is the masked spectral-grid reference (reported, not part of the window).

v_q/v_{te}	window $\omega_{pe}t$	limiter	max ledger (B)	grid agrees with B until
0.03	1000	cap $T = 1000$	6.6×10^{-12}	1000 / 1000
0.05	892	positivity, $N_x = 64$	3.3×10^{-12}	979.5 / 962.5
0.07	791.5	positivity, $N_x = 64$	9.7×10^{-13}	870 / 863

rejected: the unseeded deficit is only 1.4×10^{-6} of the seeded one. Only the seed scales 0, 0.1 and 1 at $t \leq 400$ were run.

D. Energetics and the onset of the OTSI

Why the transfer stays linear is energetic. At the window end of $v_q/v_{te} = 0.07$, 371 of the $W_{\rm ext} = 381.3\, nT_eV$ resides in coherent $k = 0$ electron quiver kinetic energy; the ions receive 0.05 % of $W_{\rm ext}$. The energy of the $k \neq 0$ electron motion is small and oscillating: with the mean flow removed it sloshes between wave kinetic and field energy with a period of about $67\,\omega_{pe}^{-1}$ near the onset, so a single time sample is not a heating measure; the average over $200\,\omega_{pe}^{-1}$ (the HHS filter, Sec. VII A) is 5 initial thermal energies at the end of the window. The fluctuating-field amplitude ($\sum_{k\geq 1} |E_k|^2$) reaches about $2077\, E_0^2$ only when the OTSI turns on, at $\omega_{pe}t = 783.5$ ($N_x = 32$) to 862 (higher N_x and N_n members; onset defined in the gates, compared within equal N_x). After that the run ends by positivity loss. Within the window the instability has not extracted a resolvable fraction of the work. *This is the main limitation of the bound*: it says that no saturation appears *before* the cascade of fluctuations starts, and nothing about what happens after.

E. Finite reservoir

In the self-consistent mode (Table V) the dark energy $U_D(t)$ follows the linear coupled-oscillator prediction at all three drives: within $0.039\, U_{D0}$ (maximum over six cases, $u = 0.1, 1, 10$) at $v_q/v_{te} = 0.01$ and 0.03 over the windows 1000 and 940, and within 0.23 % and 0.81 % of U_{D0} at $v_q/v_{te} = 0.07$ for $u = 1$ and 10 over the window 791.5. The beat times equal π/η ($\eta = 0.049$ and 0.016 for $u = 1, 10$). The accumulated work W_D follows the matched prescribed drive until the split time $t_{\rm split} = 50.5$ and 72 (equal to the linear values): the reservoir gives up 47 % of its energy for $u = 1$ and exchanges energy through beats for $u = 10$, with W_D within 1.0 % and 2.7 % of the linear prediction while it is above 20 % of its running maximum. Finite reservoir depletion is therefore captured by the linear theory; no nonlinear reservoir response is resolved. The ledger closes to 1.7×10^{-9} over all 33 runs of the strong-drive study (declared bound 10^{-6}).

F. Finite coherence

Figure 3c shows the ensemble ratio of the kinetic to the linear same-drive work. At $v_q/v_{te} = 0.01$ (window 1000, with τ_c the field coherence time) the ratios are 1.000 ± 0.007, 1.089 ± 0.064 and 0.998 ± 0.003 for $\tau_c\omega_{pe} = 30, 100, 300$; the value at $\tau_c\omega_{pe} = 100$ is 1.4 standard errors above unity and is consistent with the sampling spread of six realizations. At $v_q/v_{te} = 0.07$ the ratios are 1.0050 ± 0.0058 and 1.0000 ± 0.0038 for $\tau_c\omega_{pe} = 100$, 300, with bounds on the ensemble nonlinear correction (including twice the standard error) of 1.7 % and 0.8 %; individual realizations deviate more (up to 4.7 % and 1.5 % of the peak linear work), with ratios on both sides of unity. The late work rate saturates at about $E_0^2\tau_c\omega_L^2/4$ instead of growing, as in the linear theory. The first coherence runs realized a phase variance dt/τ_c rather than the declared $2\,dt/\tau_c$, i.e. a coherence time twice the nominal; the records were kept and relabelled and a corrected version was declared and run (Table II).

G. Scope of the reservoir and coherence bounds

The bounds hold for one box ($L = 40$), one ion mass, $T_i = T_e$, drives up to $v_q/v_{te} = 0.07$, windows set by the AW instability, fluctuation seeds of one amplitude (5×10^{-4}), a Lorentzian line only, and reservoir and coherence not combined in one run. Not done: a Gaussian line, windows beyond about 790 at 0.07, $v_q/v_{te} > 0.07$ and inhomogeneous dark fields. The grid reference (masked spectral grid) agrees with the Hermite coarse member until 870 and 863 at $v_q/v_{te} = 0.07$ (Table III).

VI. COLLISIONAL ABSORPTION

A. Linear drag law

With collisions the uniform relative velocity relaxes at the Morse rate, identical in LBO-EM and LBO-G ($\beta =$

TABLE IV. Seed-scale test of the deficit $D = W_{\rm ext}/W_{\rm lin} - 1$ at $v_q/v_{te} = 0.07$, $N_x = 32$, $N_n = 64$; the seed amplitudes are multiplied by s.

seed scale s	$D(100)$	$D(200)$	$D(300)$	$D(400)$
1	-1.03×10^{-4}	-4.15×10^{-4}	-9.35×10^{-4}	-1.66×10^{-3}
0.1	-1.03×10^{-6}	-4.15×10^{-6}	-9.36×10^{-6}	-1.66×10^{-5}
0 (unseeded)	1.00×10^{-10}	8.58×10^{-10}	1.58×10^{-9}	2.27×10^{-9}

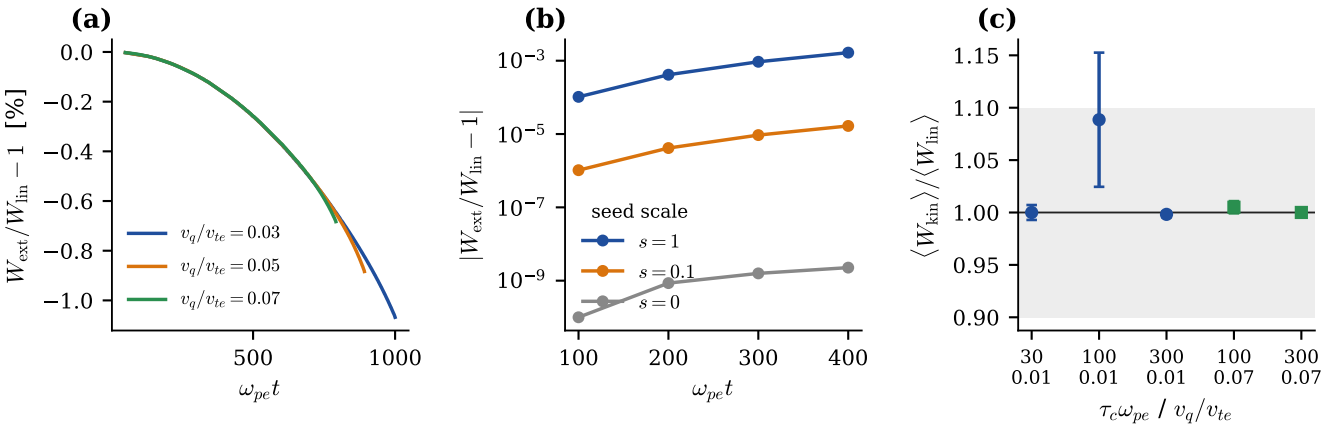


FIG. 3. Finite reservoir and coherence. (a) $W_{\rm ext}/W_{\rm lin} - 1$ for the $N_x = 32$, $N_n = 64$ member at the three drives, each to its certified window; the three curves coincide until the OTSI stage. (b) The deficit at $v_q/v_{te} = 0.07$ for seed scales $s = 1$, 0.1, 0. (c) Finite coherence: ensemble ratio $\langle W_{\rm kin}\rangle/\langle W_{\rm lin}\rangle$ with standard error over six realizations for $\tau_c\omega_{pe} = 30, 100, 300$ at $v_q/v_{te} = 0.01$ (window 1000) and $100, 300$ at $v_q/v_{te} = 0.07$ (window 791.5); band: $\pm 10\,\%$.

0),

$$\nu_{\rm eff} = \frac{\nu_{ei}}{2}\Big(1 + \frac{m_e}{m_i}\Big), \tag{11}$$

so the uniform driven cold response obeys $\dot J = \omega_L^2(\bar E + E_D) - \nu_{\rm eff} J$ and the steady absorbed power per volume is

$$P_{\rm ss} = \frac{\omega_L^2\,\nu_{\rm eff}\,\omega^2 E_0^2}{2\,[(\omega_L^2 - \omega^2)^2 + \nu_{\rm eff}^2\omega^2]}, \tag{12}$$

$$P_{\rm ss}^{\rm res} = \frac{\omega_L^2 E_0^2}{2\nu_{\rm eff}}, \tag{13}$$

with the transient decaying as $\exp(-\nu_{\rm eff}t/2)$ (1/e time $4/\nu_{ei}$). This is the cold Drude response of Dubovsky and Hernández-Chifflet [2], who write the friction as $-m_e\nu(u_e - u_i)$ with a Spitzer-like ν and find a shielding suppression $\propto m^2/\omega_p^2$ off resonance; and it matches the non-resonant absorption rate of McDermott and Witte [3]. Their ν corresponds to $\nu_{\rm eff}$ here. In our model ν is *constant*: the Spitzer scaling $\nu \propto T_e^{-3/2}$ is absent.

B. Design and gates

The set-up is that of the strong-drive study (mobile ions, pump frame, $c = 64$ with Lawson, $T = 1000$) with Dougherty e–e, e–i and i–i collisions in both conventions. The grid is $v_q/v_{te} = 0.01, 0.03, 0.05$ by $\nu_{ei} = 10^{-3}, 3\times10^{-3}, 10^{-2}, 3\times10^{-2}$ by $\omega = \omega_L, 0.9, 1.1$ (in ω_{pe}): 192 runs in all, including $N_x = 64$ and $N_n = 128$ refinement members and unseeded controls. Checks were declared before the runs. Every LBO-EM case reaches $t = 1000$ with refinement pairs agreeing to the end and a refinement spread of the late rate $\le 2.2 \times 10^{-4}$. The nonrelativistic window ($T_e/T_e(0) < 90$, i.e. $v_{te} < 0.3c$) is shorter only for $v_q/v_{te} = 0.05$ resonant. Rates are evaluated at $t_b = 996.5$.

C. Kinetic against linear

Table VI and Fig. 4 give the result. Where collisions act and the linear transient has decayed, the kinetic absorption equals the linear drag law: at $\nu_{ei} = 3 \times 10^{-2}$ the late rate is within 0.4 % of $P_{\rm ss}$ for all three frequencies and drives, and the steady-state law holds in 12 of 12 cases where $P_{\rm lin}$ is within 10 % of $P_{\rm ss}$. For the other cases the kinetic rate equals the exact *transient-inclusive* linear law to 5 % except in six cases, but the steady state is not reached within $t = 1000$: for $\nu_{ei} = 10^{-3}, 3 \times 10^{-3}, 10^{-2}$ the resonant rate is only 0.20, 0.48, 0.89 of $P_{\rm ss}$ (the amplitude transient has 1/e time $4/\nu_{ei}$). The sustained law is therefore tested directly at $\nu_{ei} = 3 \times 10^{-2}$ (and 10^{-2} detuned), and through the linear ODE for the rest. The

TABLE V. Finite reservoir (self-consistent dark field, $\Omega_D = \omega_L$), $v_q/v_{te} = 0.07$.

$u = U_{D0}/nT_eV$	η	max $\lvert U_D - U_D^{\rm lin}\rvert/U_{D0}$ [%]	$t_{\rm split}$	W_D bound [%]
1	0.049	0.23	50.5	1.0
10	0.016	0.81	72	2.7

TABLE VI. Late-time absorbed power, kinetic ($N_x = 32$, $N_n = 64$, LBO-EM) against the exact transient-inclusive linear law, $v_q/v_{te} = 0.03$. $T_e/T_e(0)$ at t_b; last column: kinetic equals linear within 5 %.

ω/ω_{pe}	ν_{ei}/ω_{pe}	$P_{\rm lin}/P_{\rm ss}$	$P_{\rm kin}/P_{\rm ss}$	$P_{\rm kin}/P_{\rm lin}$	$T_e/T_e(0)$	5 % check
res.	0.001	0.20	0.192	0.9830	35.2	pass
res.	0.003	0.48	0.472	0.9841	68.6	pass
res.	0.01	0.89	0.877	0.9908	83.6	pass
res.	0.03	1.00	0.996	0.9979	48.1	pass
0.9	0.001	2.00	2.279	1.1406	1.02	FAIL
0.9	0.003	1.54	1.584	1.0281	1.05	pass
0.9	0.01	1.11	1.106	0.9994	1.12	pass
0.9	0.03	1.00	1.003	1.0000	1.32	pass
1.1	0.001	1.38	1.496	1.0857	1.02	FAIL
1.1	0.003	1.21	1.224	1.0145	1.05	pass
1.1	0.01	1.04	1.037	0.9996	1.14	pass
1.1	0.03	1.00	1.000	1.0000	1.38	pass

5 % agreement check passes in 30 of 36 cases.

The six failures are the detuned $\nu_{ei} = 10^{-3}$ cases, where the kinetic late rate is 8–14 % high (mean +14 % at 0.9 and +9 % at 1.1), independent of v_q/v_{te}. In the unseeded controls the kinetic and linear laws agree to 3×10^{-8}, and a seed-scale follow-up (declared after the main analysis and before its runs) shows the departure scales as s^2 (ratios 0.0105 and 0.0099 for $s = 0.1$, against $s^2 = 0.01$): it is the same seed back-reaction as for the strong-drive study, large only where $W_{\rm ext}$ is itself tiny. The resonant late rate is low by 1.5–4 % at $\nu_{ei} = 10^{-3}$ (0.963 at the lowest) and by $\leq$ 0.2 % at 3×10^{-2}; at $v_q/v_{te} = 0.05$ resonant the deficit (3.7 %) is larger than at 0.01 and 0.03, but drive amplitude and heating are not separable in these data, so a few-percent heating or drive correction is allowed but not attributed.

D. Deposition law

Fitting $\ln P = a + p \ln(v_q/v_{te}) + q \ln \nu_{ei}$ over the 12 cases per frequency (standard errors from the residual scatter, so they include the transient-state curvature) gives for resonant, 0.9 and 1.1: $p = 1.99 \pm 0.10$, 2.00 ± 0.03, 2.00 ± 0.02 (kinetic), against 2 (linear, exact), and $q = -0.51 \pm 0.05$, 0.75 ± 0.02, 0.88 ± 0.01 (kinetic) against -0.52, 0.79, 0.90 (linear). The amplitude scaling is quadratic. The ν_{ei} scaling follows the transient-inclusive linear law, not a pure power: the detuned rate grows with ν (off-resonant absorption proportional to ν, until the beat transient decays), and the resonant rate falls with ν with an exponent near $-1/2$ because the steady state is not reached. The geometric-mean correction of kinetic to linear power is $\kappa = 0.985 \pm 0.003$, 1.040 ± 0.017, 1.025 ± 0.011. The domain of the law is $v_q/v_{te} \leq 0.05$, $\nu_{ei} \in [10^{-3}, 3 \times 10^{-2}]\omega_{pe}$, $\omega \in \{0.9, \omega_L, 1.1\}\omega_{pe}$, $t \leq 1000$, one box and mass ratio, one seed level.

E. Collision form and heating

EM and G ($\beta = 0$) agree in $W_{\rm ext}$ to 2.5×10^{-6} and the late-rate ratio differs by $\leq 7 \times 10^{-6}$ (agreement checks pass in 36 of 36 cases); electron kinetic energy and temperature differ by $\leq 8 \times 10^{-3}$. Ion temperature differs by up to 31 % and the electron–ion energy exchange is halved, as expected: the absorption is set by the momentum friction, identical in the two forms, while the exchange is the temperature-difference rate. The ion-heating check of the earlier pilot failed as declared for this reason (Table II). Electron heating is large: $T_e/T_e(0)$ reaches 231 (Fig. 4d), beyond $v_{te} = 0.3c$ for $T_e/T_e(0) > 90$, i.e. outside the nonrelativistic model. Heating does not change the rate in this model because ν is constant ($R_{\rm late}$ stays within 1 % of unity while $T_e/T_e(0)$ rises from 7 to 70 at $v_q/v_{te} = 0.03$, $\nu_{ei} = 10^{-2}$, resonant). Ion heating is small ($T_i/T_i(0) \leq 2.8$) and the $e \to i$ exchange is 4×10^{-5} to 1.6×10^{-2} of $W_{\rm ext}$, growing in proportion to ν_{ei}. The steady-state self-check of the linear theory had a declaration defect; its declared follow-up passes (Table II).

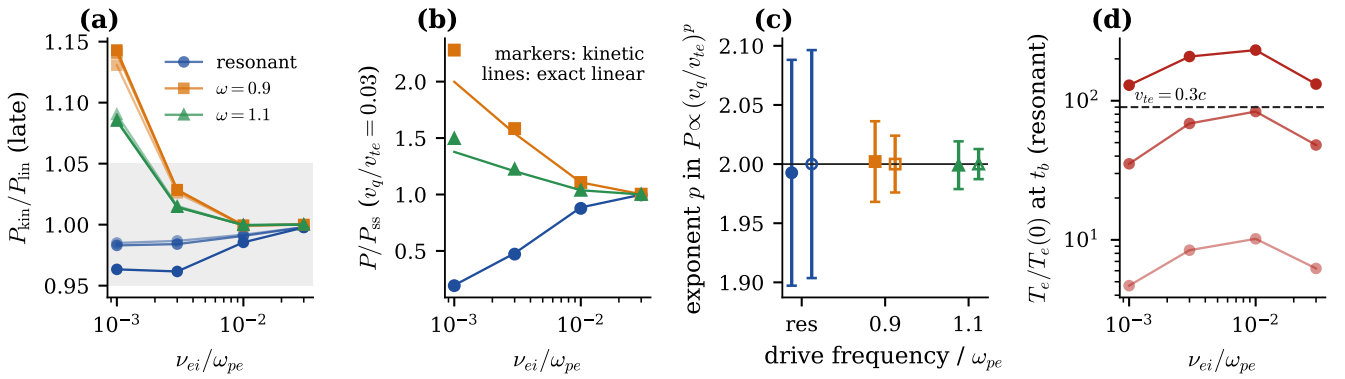


FIG. 4. Collisional absorption. (a) Ratio of the kinetic to the exact linear late-time power for $v_q/v_{te} = 0.01, 0.03, 0.05$ (increasing opacity) and three drive frequencies; band: the 5 % gate. (b) Late power relative to the steady state $P_{\rm ss}$ at $v_q/v_{te} = 0.03$, kinetic (markers) and linear (lines). (c) Fitted exponent p of the drive amplitude, kinetic (filled) and linear (open), with standard errors; the solid line marks $p = 2$. (d) Electron heating at t_b for the resonant cases, same opacity code as (a); dashed: $v_{te} = 0.3c$, beyond which the nonrelativistic model is outside its validity.

F. What is needed for a cosmological statement

Our collisions fix ν at a constant. A cosmological deposition law needs (i) the velocity-dependent Coulomb frequency $\nu(T_e) \propto T_e^{-3/2}$, which couples the heating to the absorption and lowers it; (ii) radiative and Compton cooling; (iii) cosmological expansion and ionization history; (iv) relativistic electrons, since $T_e/T_e(0)$ exceeds the nonrelativistic limit; (v) the $k \neq 0$ coupling of a real dark-photon field; and (vi) a steady-state measurement at $\nu_{ei} \leq 10^{-2}$ resonant, which needs $t \gg 4/\nu_{ei}$. ν_{ee} and ν_{ii} are order-of-magnitude choices that do not enter the uniform drag.

VII. DISCUSSION

What the results do establish. Within a one-dimensional, nonrelativistic, deterministic-seed model with the ingredients of HHS, and for the drives and times listed in Table III, the energy transfer from a uniform resonant drive follows the linear resonant law: $v_q/v_{te} = 0.03$ to $\omega_{pe}t = 1000$, 0.05 to 892 and 0.07 to 791.5. At the strongest drive the nonlinear correction to $W_{\rm ext}$ is bounded by 0.68 %, and the work has reached $W_{\rm ext} = 381.3\, nT_eV$ (111, 247 and 381 nT_eV at the window ends for 0.03, 0.05, 0.07), almost all of it coherent $k = 0$ quiver energy. A finite reservoir and a finite coherence time of $\tau_c\omega_{pe} \geq 100$ do not change this. Collisions give the linear Drude law, with the kinetic rate equal to the linear law to a few per cent and to 0.4 % at steady state.

The nonrelativistic limitation. The model is nonrelativistic: the Vlasov equation uses the Newtonian momentum and the kinetic-energy ledger the quadratic form. This matters for the quantities reported here. The coherent $k = 0$ quiver energy at the window end, $W_{\rm ext} \simeq 111$, 247 and 381 nT_eV for $v_q/v_{te} = 0.03$, 0.05, 0.07, corresponds to an electron speed $\sqrt{2W_{\rm ext}/(nm_e)}$ of $0.47c$, $0.70c$ and $0.87c$ (nonrelativistic formula), so that for most of every window the quiver is beyond the range in which Newtonian dynamics is accurate, and the electron temperature in the collisional runs reaches $v_{te} > 0.3c$ (Fig. 4d). Relativistic mass increase lowers the plasma frequency of the quivering electrons by a fraction of the order of v^2/c^2, which for $v \gtrsim 0.2c$ exceeds the resonance width $\sim 1/(\omega_{pe}t)$ of the driven oscillator long before the end of our windows. A relativistic model would therefore detune the resonance, and could saturate the transfer by that mechanism alone, independently of the nonlinear plasma processes that HHS invoke. Our linear-law result is thus a statement about the nonrelativistic model; it is not a statement about the relativistic plasma, and it cannot be compared with a relativistic PIC run beyond the time at which the quiver speed is small ($\omega_{pe}t \lesssim 200$ at $v_q/v_{te} = 0.03$). We have not run a relativistic model, and this is the largest untested difference from HHS (Sec. VII A).

What they do not establish. (i) The growth of higher-k fluctuations by the strong pump, which in our model is the OTSI (we take this to be the process described in the HHS abstract; that identification is our reading, not something we tested against their simulations), has its onset at $\omega_{pe}t \simeq 783.5$–862 at $v_q/v_{te} = 0.07$, at the end of our window. The windows end by positivity loss at nearly the same times, because the numerical instability of the Hermite basis reaches the resolved range there, so we cannot say whether the transfer would saturate after the cascade develops. The bound is a statement about the interval before it. In our runs about 97 % of the transferred energy is still coherent $k = 0$ quiver energy at the window end. HHS do not count that coherent part as deposited (their Figs. 2 and 3 show the $k \neq 0$ kinetic energy), so the large $W_{\rm ext}$ is not com-

parable to their saturation level; what is comparable is stated in Sec. VII A. For the claim at times after the cascade, ours is neither a confirmation nor a refutation. (ii) The model is one-dimensional (and nonrelativistic, above), with a box $L = 40\,c/\omega_{pe}$ and the strict 2/3 mask retaining Fourier modes $k \le 9$ at $N_x = 32$ and $k \le 20$ at $N_x = 64$ (in units of $2\pi/L$), against 1000 cells in HHS; the retained modes cover the onset but not a developed spectrum. (iii) The seed amplitude is imposed; at this seed the deficit is 0.68 %, and a different fluctuation level rescales it. (iv) Drives above $v_q/v_{te} = 0.07$ (including $v_q/v_{te} = 0.1$) have no window beyond about 760, so the claims are bounds for $v_q/v_{te} \le 0.07$ and $\omega_{pe}t \lesssim 790$ at the strongest drive, and nothing more.

Why a nonlinear detuning could be expected. The trapping time of O'Neil [20], $\tau = (m/eEk)^{1/2}$, ends linear theory, and the ponderomotive coupling is the origin of the modulational and collapse processes of Zakharov [21]. For the uniform pump the mean-field law is exact and neither matters until finite-k fluctuations grow, which is why the first deviations in our model are those of the finite-k OTSI.

A. Comparison with the HHS simulations

What HHS plot. The energy "$\Delta E/n_e m_e$" of their Figs. 2 and 3 is the change of the thermal energy density $n_\alpha m_\alpha v_{th}^2/2$ of each species (SM Eqs. S35 and S36), with v_{th}^2 the one-dimensional (x) velocity variance about the mean; SM footnote 8 states that the plotted temperatures measure the total kinetic energy of electrons and ions with the $k = 0$ component removed, and that energy already thermalized cannot be told apart from energy in $k \neq 0$ oscillations. The coherent $k = 0$ quiver energy is therefore *not* counted as deposited; the mean field enters separately as $\langle E^2\rangle/2$ (red solid line). The temperature is a global (box) variance, not a local one. All curves are filtered with a moving average over $200\,\omega_{pe}^{-1}$ (Fig. 2 caption). Their statement that the deposited energy is about the initial thermal energy ($5 \times 10^{-4}\, n_e m_e c^2$ for $v_{th}^2 = 10^{-3}c^2$) is a statement about this $k \neq 0$ energy. In Sec. II of the SM the particle number per cell is raised for weak drives so that the shot-noise energy equals the driven energy at $\omega_{pe}t \approx 2.4$ (SM Eq. S38), and the driven energy is converted into incoherent $k \neq 0$ noise within $\omega_{pe}t \simeq 3$–4 (SM Sec. II).

Same definition here. Our records carry the box-mean-flow-subtracted one-dimensional electron variance, $\Delta E = \frac{1}{2} m_e \Delta(M_{xx} - M_x^2/n)$, and the mean field $\bar{E}$; the figure script evaluates both with the trailing $200\,\omega_{pe}^{-1}$ average. The definition is therefore the same as theirs, one-dimensional, global, with the $k = 0$ mean flow removed.

What is comparable. Only the resonant $v_q/v_{te} = 0.03$ case is: same drive, T_e, mass ratio, box, filter and definition, and our window ($\omega_{pe}t \le 1000$) covers the time over which their top panel of Fig. 2 saturates ($\omega_{pe}t \lesssim 500$; SM Sec. III notes the loss of resonance at about 800). Table VII compares the numbers; the HHS entries are read off their logarithmic plot (accuracy about a factor of 1.5). Their weak case, $v_q/v_{te} = 10^{-3}$, saturates at $\omega_{pe}t \simeq 1500$–2300, beyond our windows, and we have no run at that drive; their Landau–Zener runs (Fig. 3) and their T_e/T_i histories to $\omega_{pe}t = 8 \times 10^4$ (SM Fig. S3) have no counterpart here; our $v_q/v_{te} = 0.05$ and 0.07 are not drives they show, and we did not run those in their model.

Reading. At this matched drive and over the time in which HHS see the $k \neq 0$ energy rise to about fifty initial thermal energies and the field energy collapse, our electron $k \neq 0$ energy is 0.1 initial thermal energies at $\omega_{pe}t = 500$ and 2.5 at 1000, and the mean-field energy follows the linear law to 0.96 of it. We therefore do not reproduce their saturation at $v_q/v_{te} = 0.03$. We do not conclude that their result is wrong; the model differs in ways we have not tested: (a) the loading and the seed level (our deterministic seed has relative amplitude 5×10^{-4}; theirs is shot noise set by the particle number, with the noise equal to the driven energy at $\omega_{pe}t \approx 2.4$); (b) our Fourier truncation retains $k \le 1.4\,\omega_{pe}/c$ at $N_x = 32$ and ≤ 3.1 at $N_x = 64$, i.e. $k\lambda_{De} \le 0.045$ and 0.1, whereas their instabilities populate wavenumbers up to λ_{De}^{-1} (SM Sec. III); (c) their pusher is relativistic, and the accumulated quiver speed $v_q^D \omega_{pe}t/2$ reaches about $0.1c$ by $\omega_{pe}t \approx 200$ and $0.2c$ by 500 ($0.47c$ at 1000 from our work ledger), so that the relativistic frequency shift, of the order of v^2/c^2, can exceed the resonance width $\sim 1/t$ well before their saturation time; our model has no such shift. We regard (c) as the most likely single cause of the difference at $v_q/v_{te} = 0.03$, but this is a hypothesis: we have not run a relativistic model, and items (b) and (c) can only be settled by runs we have not done. Our numbers at $v_q/v_{te} = 0.05$ and 0.07 are bounds in this model and are not comparable to any HHS figure.

VIII. CONCLUSIONS

We built and verified a noise-free Hermite–Fourier solver for the canonical Maxwell–Proca–Vlasov system. Its limit is a numerical truncation instability of the asymmetrically weighted Hermite basis, which bounds certified windows to 1000, 892 and 791.5 at $v_q/v_{te} = 0.03$, 0.05, 0.07; no higher resolution, stronger closure, or alternative basis we tried (Dai, symmetric weight, square-root form) removes it. Inside those windows, the energy transfer follows the linear resonant law to 0.68 %, with no resolved frequency detuning, finite dark-reservoir depletion and finite coherence follow the linear predictions, and the small deficit is seed back-reaction. With collisions, the absorption is the linear Drude law with $\nu_{\rm eff} = (\nu_{ei}/2)(1 + m_e/m_i)$, quadratic in the drive, and independent of the collision form. These are bounds in a restricted model. They do not test nonlinear suppression after the oscillating two-stream cascade develops, relativistic electrons (the coherent quiver speed reaches

TABLE VII. Resonant drive $v_q/v_{te} = 0.03$: HHS (PRL Fig. 2, top; values read off the plot) against this work ($N_x = 32$, $N_n = 64$). ΔE_e: change of the one-dimensional $k \neq 0$ electron kinetic energy; $\bar{E}^2/2$: mean-field energy ($\langle E^2\rangle/2$ for HHS); both in units of $n_e m_e c^2$ with the $200\,\omega_{pe}^{-1}$ trailing average; $E_{\rm th} = 5 \times 10^{-4}$.

	HHS	ours, $\omega_{pe}t = 500$	ours, $\omega_{pe}t = 1000$
ΔE_e	$\approx 2.5 \times 10^{-2}$ ($\approx 50\,E_{\rm th}$)	5.0×10^{-5} ($0.1\,E_{\rm th}$)	1.2×10^{-3} ($2.5\,E_{\rm th}$)
$\bar{E}^2/2$	$\approx 10^{-3}$ at 800, $\lesssim 10^{-4}$ at 1000	9.1×10^{-3} (linear 9.2×10^{-3})	4.4×10^{-2} (linear 4.6×10^{-2})

$0.47c$ at $v_q/v_{te} = 0.03$ and $0.87c$ at $v_q/v_{te} = 0.07$ within the windows, so the relativistic detuning of the resonance is the main open difference from HHS), $k \neq 0$ coupling, or a temperature-dependent collision frequency; a stable basis is needed to extend the windows.

DATA AND CODE AVAILABILITY

All code, records and generated results are in the Dark-SPECTRAX repository [22] (`https://github.com/uwplasma/Dark-SPECTRAX`). The records and study scripts are those of commit `a3ef7a6` of the `main` branch; the manuscript, figures and number macros are in `paper/` at the commit that follows it (stated in `paper/README.md`). No release or tag has been made. Figures, tables and quoted numbers are generated from the records by `paper/make_figures.py`; the record index below maps each figure and table to its files. The parent solver SPECTRAX [6] was used at the commits stored in the records: `3de6b2b` (finite reservoir and coherence), `4090fbe` (strong-drive study) and `35231ab` (collisions; the pin in `pyproject.toml`); the branch tip at the time of writing is `729d7fe`. These commits belong to the `integration/dark-baseline` branch of SPECTRAX, an unmerged integration branch that collects open parent pull requests with the features used here (field-scaled closure, Lawson factor, collision conventions); it is not a release. Checks were committed in `studies/r1_common.py`, `r1nl_gates.py` and `r2_gates.py` before the runs.

Record index (paths relative to the repository root). Fig. 1 and Table I: `docs/_static/b00/run.json`, `studies/instab/otsi.json`, `studies/c05/run.json`, `studies/gpures/c08.json`, `studies/c09/run.json` (summary in `docs/results.md`). Fig. 2: `studies/highk/eig.json`, `studies/gpures/summary.json`, `studies/rclosure/stagef.json`. Tables III and V, Fig. 3a: `studies/r1nl/stage1_summary.json` and `stage2_summary.json`. Table IV and Fig. 3b: `studies/r1nl/followup_summary.json`. Fig. 3c and the weak-drive reservoir runs: `studies/r1/coherence2_summary.json`, `studies/r1/reservoir_summary.json`. Table VI and Fig. 4: `studies/r2/summary.json`. Table VII: `studies/r1nl/runs`. Failed and amended checks (Table II): `docs/status.md`.

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